

Research Landscape Brief: Graphene research and applications

A cited, primary-source snapshot of research momentum, leading players, and must-read work. Every figure is computed from real OpenAlex records and linked so you can verify it.

Sample corpus: 263 works · Data: OpenAlex (CC0) · Generated July 21, 2026

1. Research momentum (works per year in this sample)

Year	Works	
2014	10	■■■■■
2015	10	■■■■■
2016	3	■
2017	4	■■
2018	3	■
2019	6	■■■
2020	1	■
2021	1	■

2. Leading institutions & organizations (by output)

- University of Manchester — 32 works
- The University of Texas at Austin — 22 works
- Radboud University Nijmegen — 15 works
- Columbia University — 15 works
- Centre National de la Recherche Scientifique — 12 works
- Massachusetts Institute of Technology — 12 works
- University of Cambridge — 11 works
- Chinese Academy of Sciences — 11 works

3. Active subtopics / keywords

Materials science (241), Graphene (240), Nanotechnology (228), Physics (124), Chemistry (107), Composite material (102), Condensed matter physics (88), Optoelectronics (84), Graphite (73), Oxide (63)

4. Most-cited work (verifiable via DOI)

Electric Field Effect in Atomically Thin Carbon Films (2004) — 66,329 citations

Verify: <https://doi.org/10.1126/science.1102896>

The rise of graphene (2007) — 39,569 citations

Verify: <https://doi.org/10.1038/nmat1849>

The electronic properties of graphene (2009) — 24,689 citations

Verify: <https://doi.org/10.1103/revmodphys.81.109>

Two-dimensional gas of massless Dirac fermions in graphene (2005) — 21,482 citations

Verify: <https://doi.org/10.1038/nature04233>

Measurement of the Elastic Properties and Intrinsic Strength of Monolayer Graphene (2008) — 20,725 citations

Verify: <https://doi.org/10.1126/science.1157996>

5. Geographic concentration

US (738), GB (269), CN (129), KR (81), JP (68), NL (62)

Method & limits

Computed from a live OpenAlex sample (CC0 public data). This free sample uses a partial corpus for speed; a paid brief runs the full topic set, adds company-vs-peer positioning, funder analysis, and a 5-year trend. We never assert commercial outcomes — only what the publication record shows.

Want this for your exact topic or competitor set? Reply and we'll scope a full cited brief.